

Year 7 · Chapter 3 · Whole numbers: answers (teacher copy)

3.01 Rounding and estimation (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: 88 000
- Ex 1b: 87 530
- Ex 2a: $400 + 80 + 900 + 20 = 1400$
- Ex 2b: $900 + 300 - 400 = 800$
- Ex 3a: $90 \times 20 = 1800$
- Ex 3b: $2400 \div 30 = 80$

Easy · Set A (quick drill)

- 50
- 150
- 3 880
- 1 000
- 6 000
- 87 530
- 300
- 1 300
- 8 000
- 15 000
- 100 000
- 600

Easy · Set B

- 700
- 87 500
- 3 000
- 90 000
- 65 000
- 150 000

Medium · Set C

- $600 + 300 = 900$
- $4000 - 2000 = 2000$
- $40 + 70 + 400 = 510$
- $800 - 200 + 100 = 700$
- $7000 + 3000 = 10\,000$
- $5000 - 4000 = 1000$

Challenging · Set D

- $50 \times 20 = 1000$ (exact 1008)
- $600 \times 9 = 5400$ (exact 5508)
- $4000 \div 40 = 100$ (exact ≈ 103.9)
- $30 \times 30 = 900$ (exact 868)

Extension

- 11 500 and 12 499
- 35 499

Exit ticket

- Level 1: 6400
- Level 2: $500 + 300 - 100 = 700$
- Level 3: $4000 \div 20 = 200$

3.02 Multiplication (booklet p. 7)

WE DO examples (teacher)

- Ex 1a: 48
- Ex 1b: 200
- Ex 2a: 170
- Ex 2b: 297
- Ex 3a: 882
- Ex 3b: 10 170

Easy · Set A (quick drill)

- 48
- 90
- 180
- 88
- 100
- 350
- 56
- 54
- 60
- 120
- 132
- 150
- 420
- 1 000

Easy · Set B

- 170
- 200
- 297
- 80
- 180
- 231

Medium · Set C

- 882
- 832
- 2 250
- 744
- 3 702
- 4 131

Challenging · Set D

- 10 170
- 1 081
- 14 040
- 1 368 people

Extension

- 3 ($37 \times 5 = 185$)
- e.g. 25×40 or 20×50

Exit ticket

- Level 1: 70
- Level 2: 1 458
- Level 3: 3 456

3.03 Division (booklet p. 11)

WE DO examples (teacher)

- Ex 1a: 240
- Ex 1b: 16
- Ex 2a: 124
- Ex 2b: 132
- Ex 3a: 162
- Ex 3b: 121

Easy · Set A (quick drill)

1. 240
2. 8
3. 9
4. 35
5. 12
6. 7
7. 8
8. 9
9. 12
10. 45
11. 12
12. 12
13. 16
14. 50

Easy · Set B

1. 16
2. 18
3. 30
4. 32
5. 81
6. 12

Medium · Set C

1. 124
2. 132
3. 162
4. 121
5. 235
6. 156

Challenging · Set D

1. 79 r 1
2. 250 r 3
3. 714 r 2
4. 6 buses ($250 \div 45 = 5 \text{ r } 25$)

Extension

1. 26 or 38
2. 6

Exit ticket

1. Level 1: 36
2. Level 2: 123
3. Level 3: 156 r 2

3.04 Divisibility tests (booklet p. 15)**WE DO examples (teacher)**

1. Ex 1a: yes; no; no
2. Ex 1b: no (digit sum 11); no
3. Ex 2a: yes (08); yes ($208 \div 8 = 26$)
4. Ex 2b: no (not divisible by 3)
5. Ex 3a: B: 24
6. Ex 3b: 2 (522)

Easy · Set A (quick drill)

1. yes
2. no
3. yes
4. yes
5. no
6. yes
7. 12, yes
8. 1, no
9. 12, yes
10. 18, yes
11. 11, no
12. 27, yes
13. 18, yes
14. 10, no

Easy · Set B

1. yes (12)
2. no
3. yes (18)
4. no (1)
5. yes (12)
6. yes (18)

Medium · Set C

1. by 4, 6
2. by 4, 8
3. by 4, 6
4. by 4, 8
5. by 6
6. by 4, 8

Challenging · Set D

1. by 2, 3, 4, 5, 6, 8, 9, 10
2. 24, 96, 120
3. 1, 4, 7
4. No: $9 = 3 \times 3$ so anything divisible by 9 is divisible by 3

Extension

1. Yes: ends in 0 (by 5) and digit sum 18 (by 3)
2. 1020

Exit ticket

1. Level 1: yes
2. Level 2: yes (digit sum 18)
3. Level 3: yes: even and digit sum 15

3.05 Long division (booklet p. 19)**WE DO examples (teacher)**

1. Ex 1a: 26
2. Ex 1b: 124
3. Ex 2a: 256 r 7
4. Ex 2b: 54
5. Ex 3a: 156 r 8
6. Ex 3b: 23 buses

Easy · Set A (quick drill)

1. 84
2. 150
3. 96
4. 120
5. 96
6. 144
7. 52
8. 84
9. 69
10. 105
11. 72
12. 51
13. 88
14. 95

Easy · Set B

1. 8
2. 7
3. 6
4. 16
5. 8
6. 9

Medium · Set C

1. 26
2. 124
3. 54
4. 112
5. 255
6. 234

Challenging · Set D

1. 256 r 7
2. 156 r 8
3. 470 r 6
4. 23 buses

Extension

1. 994
2. 993

Exit ticket

1. Level 1: 6
2. Level 2: 108
3. Level 3: 148 r 4

3.06 Roman numerals (booklet p. 23)

WE DO examples (teacher)

1. Ex 1a: 37
2. Ex 1b: 240
3. Ex 2a: 24
4. Ex 2b: XXVIII
5. Ex 3a: LXXIV
6. Ex 3b: CXXV

Easy · Set A (quick drill)

1. 8
2. 12
3. 37
4. 60
5. 210
6. 15
7. III
8. IX
9. XIV
10. XIX
11. XL
12. LII
13. XCIX
14. CL

Easy · Set B

1. 4
2. 9
3. 24
4. 40
5. 240
6. 90

Medium · Set C

1. VI
2. XIII
3. XXVIII
4. LXXIV
5. CXXV
6. XLIX

Challenging · Set D

1. MCMXCIX
2. 2026
3. e.g. 2026 = MMXXVI
4. Only I before V or X is allowed (99 = XCIX)

Extension

1. CCLV
2. CVIII

Exit ticket

1. Level 1: 16
2. Level 2: XXXIX
3. Level 3: MMXXIV

3.07 Index notation and powers (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: 3^4
2. Ex 1b: 7^7
3. Ex 2a: 36
4. Ex 2b: 125
5. Ex 3a: 100 000
6. Ex 3b: 256

Easy · Set A (quick drill)

1. 4^3
2. 6^5
3. 9^2
4. 2^6
5. 10^4
6. $5^3 \times 3^2$
7. 8
8. 9
9. 25
10. 1000
11. 16
12. 32
13. 1
14. 27

Easy · Set B

1. $8 \times 8 \times 8$
2. $3 \times 3 \times 3 \times 3 \times 3$
3. 11×11
4. $2 \times 2 \times 2 \times 2$
5. 7
6. $2 \times 2 \times 2 \times 5 \times 5$

Medium · Set C

1. 36
2. 125
3. 32
4. 10 000
5. 81
6. 1

Challenging · Set D

1. 256
2. $64 - 27 = 37$
3. $8 \times 9 = 72$
4. $2^5 = 32$ is larger than $5^2 = 25$

Extension

1. $3^4 = 9^2$
2. $n = 10$

Exit ticket

1. Level 1: 7^3
2. Level 2: 64
3. Level 3: $9 \times 16 = 144$

3.08 Square roots and cube roots (booklet p.

31)

WE DO examples (teacher)

1. Ex 1a: 7
2. Ex 1b: 12
3. Ex 2a: 20
4. Ex 2b: 9
5. Ex 3a: $\sqrt{144} = 12$
6. Ex 3b: 7 and 8

Easy · Set A (quick drill)

1. 4
2. 7
3. 9
4. 10
5. 12
6. 1
7. 5
8. 8
9. 11
10. 20
11. 3
12. 4
13. 5
14. 10

Easy · Set B

1. 2
2. 3
3. 4
4. 5
5. 10
6. 1

Medium · Set C

1. 20
2. 20
3. 9
4. 15
5. $6 + 8 = 14$
6. $6 \times 2 = 12$

Challenging · Set D

1. 9 and 10
2. 4 and 5
3. $\sqrt{100} = 10$; no ($6 + 8 = 14$)
4. 14 cm

Extension

1. 70 and 30
2. 7 cm

Exit ticket

1. Level 1: 8
2. Level 2: 5
3. Level 3: $11 - 10 = 1$

3.09 Prime and composite numbers (booklet

p. 35)

WE DO examples (teacher)

1. Ex 1a: 1, 2, 3, 4, 6, 12; 1, 13
2. Ex 1b: neither
3. Ex 2a: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29
4. Ex 2b: No: $51 = 3 \times 17$
5. Ex 3a: 29, 31, 37, 41, 43, 47
6. Ex 3b: 46, 48, 49, 50, 51, 52, 54, 55, 56, 57, 58

Easy · Set A (quick drill)

1. 1, 2, 3, 6
2. 1, 11
3. 1, 3, 5, 15
4. 1, 2, 4, 5, 10, 20
5. 1, 17
6. 1, 2, 3, 4, 6, 9, 12, 18, 36
7. P
8. C
9. P
10. C
11. P
12. C
13. C
14. P

Easy · Set B

1. prime
2. composite
3. prime
4. composite
5. prime
6. composite

Medium · Set C

1. 53, 59, 61, 67
2. 21, 22, 24, 25, 26, 27, 28, 30
3. 42, 44, 45, 46, 48, 49, 50, 51, 52, 54, 55, 56, 57, 58
4. 49, 51, 55, 57, 63, 65, 69

Challenging · Set D

1. every other even number has 2 as a factor
2. e.g. $5 + 31$, $7 + 29$, $13 + 23$, $17 + 19$
3. 3,5; 5,7; 11,13; 17,19; 29,31; 41,43
4. yes, 31

Extension

1. yes (not divisible by 2, 3, 5, 7, 11, 13)
2. no: 13×17

Exit ticket

1. Level 1: prime
2. Level 2: 1, 2, 4, 7, 14, 28
3. Level 3: 31, 37, 41, 43

3.10 Prime factors (booklet p. 39)

WE DO examples (teacher)

1. Ex 1a: $2^2 \times 3 \times 5$
2. Ex 1b: $2^3 \times 5^2$
3. Ex 2a: $2^4 \times 3^2$
4. Ex 2b: $2 \times 3^2 \times 5$
5. Ex 3a: 2^8 , so $2^4 = 16$
6. Ex 3b: $2^3 \times 3^6$, so $2 \times 3^2 = 18$

Easy · Set A

1. $2^2 \times 3$
2. 2×3^2
3. $2 \times 3 \times 5$
4. $2^3 \times 5$
5. $3^2 \times 5$
6. 2×5^2

Easy · Set A2 (quick drill)

1. 2×3
2. 3×5
3. 3×7
4. 5×7
5. 2×11
6. 3×13
7. 2×13
8. 7×11

Easy · Set B

1. $2^3 \times 3$
2. $2^2 \times 3^2$
3. $2^3 \times 3^2$

Medium · Set C

1. $2^2 \times 3 \times 5$
2. $2^3 \times 5^2$
3. $2^4 \times 3^2$
4. $2^2 \times 3^2 \times 7$
5. $3^3 \times 5^2$
6. $2^3 \times 7^2$

Challenging · Set D

1. $2^8 \rightarrow 16$
2. $2^2 \times 3^4 \rightarrow 18$
3. $2^3 \times 3^6 \rightarrow 18$
4. $2^3 \times 7^3 \rightarrow 14$

Extension

1. 5
2. 12

Exit ticket

1. Level 1: $2^2 \times 5$
2. Level 2: $2^2 \times 3 \times 7$
3. Level 3: $2^6 \times 3^2 \rightarrow 24$

3.11 Highest common factor (booklet p. 45)

WE DO examples (teacher)

1. Ex 1a: 6
2. Ex 1b: 4
3. Ex 2a: 12
4. Ex 2b: 17
5. Ex 3a: 81
6. Ex 3b: 36

Easy · Set A (quick drill)

1. 3
2. 4
3. 5
4. 7
5. 8
6. 1
7. 2
8. 3
9. 5
10. 4
11. 6
12. 7
13. 5
14. 6

Easy · Set B

1. 4
2. 12
3. 15
4. 8
5. 18
6. 14

Medium · Set C

1. 81
2. 40
3. 48
4. 42
5. 75
6. 14

Challenging · Set D

1. $\frac{3}{7}$
2. 8
3. 15 cm
4. e.g. 24 and 30, or 30 and 36

Extension

1. 24
2. 15

Exit ticket

1. Level 1: 4
2. Level 2: 14
3. Level 3: 84

3.12 Lowest common multiple (booklet p. 49)**WE DO examples (teacher)**

1. Ex 1a: 12
2. Ex 1b: 24
3. Ex 2a: 90
4. Ex 2b: 35
5. Ex 3a: 400
6. Ex 3b: 180

Easy · Set A (quick drill)

1. 3, 6, 9, 12, 15, 18
2. 5, 10, 15, 20, 25, 30
3. 8, 16, 24, 32, 40, 48
4. 12, 24, 36, 48, 60, 72
5. 9, 18, 27, 36, 45, 54
6. 15, 30, 45, 60, 75, 90
7. 10
8. 12
9. 20
10. 18
11. 40
12. 30
13. 24
14. 36

Easy · Set B

1. 6
2. 20
3. 18
4. 21
5. 24
6. 30

Medium · Set C

1. 90
2. 400
3. 180
4. 42
5. 72
6. 180

Challenging · Set D

1. 7:00 am (60 minutes)
2. 24
3. 4 packs of hot dogs, 5 packs of buns (40 each)
4. 3, 6, 12, 24

Extension

1. 72
2. 60 seconds

Exit ticket

1. Level 1: 12
2. Level 2: 48
3. Level 3: 84